1 Causal Inference Objectives

Distinguishing between **experimentally-generated** and **observational data** to evaluate causal claims. Key goals include fitting regression models that adjust for **confounding variables** and applying the principle: `block what you can, randomize what you cannot` in experimental designs like A/B testing.

Relevance to Data Science This block emphasizes **"joined-up thinking"** across the data pipeline. It covers randomized studies, such as **A/B testing** for website optimization, and non-randomized studies, such as assessing drug efficacy in large healthcare databases.

2 Hypothesis Testing Review

Decision	H_0 **is True**	H_0 **is False**
Fail to Reject	Correct (TN)	Type II Error (FN / β)
Reject H_0	Type I Error (FP / α)	Correct (TP / Power)

(H_0) represents the status quo in our case study, whereas (H_a) represents our hypothesis of interest.

The **p-value** is the probability of obtaining a test statistic as extreme as observed, assuming H_0 is true. Reject H_0 if `p-value < alpha`. The significance level α will be our tolerance to type I error: rejecting the null hypothesis when in fact is true.

Increasing Power $(1-\beta)$**:** Power is the ability to detect a real effect when it exists (reducing Type II errors). Increasing power helps you find true positives, but it doesn't stop "fake" positives from appearing due to multiple testing.

```
hypothesis.test <- function(alpha = 0.05, p_value) {
  reject <- (p_value < alpha)
  return(reject)}
```

2.1 P-Hacking & Type 1 Error Inflation

p-hacking is conducting many tests and only reporting significant results ($p < 0.05$). This inflates Type I error. For m independent tests, the probability of at least one Type I error is $1-(1-\alpha)^m$.

3 Bonferroni Correction

3.1 Family-Wise Error Rate (FWER)

Sets a stricter significance level: `alpha / m`. It ensures the **FWER** (chance of > 1 false rejection) is $< \alpha$.

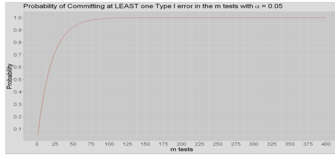
```
sum(pval < (sig_alpha / (nvars * nvars))) # Number of rejected H_0

p.adjust(pval, method = "bonferroni") # Bonferroni Adjustment in R
```

3.2 Trade-offs

Bonferroni is **highly conservative**. It significantly increases the risk of **Type II errors** (missing real effects).

```
fdr_p <- p.adjust(raw_p, method = "fdr")# FDR/BH Adjustment in R
```



4 False Discovery Rate (FDR)

Controls the **expected proportion of false positives** among all rejections. The **Benjamini-Hochberg (BH)** method ranks m p-values ($p_{(1)} \leq \cdots \leq p_{(m)}$) and finds the largest k such that $p_{(k)} \leq \frac{k}{m}\alpha$.

4.1 Selecting a Correction

Bonferroni:if you prefer to be very conservative and would rather miss a real effect than report a fake one. **FDR**:if you prefer to be less conservative and want to ensure you don't miss important discoveries, even if it means a small, known percentage of your results might be false positives.

5 Simpson's Paradox

5.1 Definition & Mechanics

Simpson's paradox occurs when a trend appearing in aggregated data reverses when the data is separated into specific groups. This often results from a limited or poor Exploratory Data Analysis (EDA). A real-world example is COVID-19 fatality rates: Italy appeared to have a higher overall rate than China, but when disaggregated by age, Italy actually had lower rates in every single age group.

6 Confounding Variables

6.1 Criteria for Confounding

A variable C is a confounder of the relationship between exposure X and outcome Y if:

- It must be related to the outcome (prognosis or susceptibility).
- Its distribution must be different in the groups being compared.

Confounding is essentially a mixing of effects where the third factor independently affects the risk of the outcome.

```
# Simulating Simpson's Paradox
simpson_data <- simulate_simpson(n=200, groups=4, r=0.4)

# Overall vs Group correlation
cor(data$X, data$Y) # May be negative
data %>% group_by(Group) %>%
  summarise(corr = cor(X, Y)) # Positive
```

7 Dealing with Confounding

7.1 Observational vs. Experimental

Observational Studies: Use stratification or regression adjustment. Include confounders C as additional regressors to "block" their effect.

Experimental Studies (A/B Testing): Randomization is the "gold standard." It guarantees that treatment X is independent of all confounders, allowing for causal interpretation.

8 TikTok Simulation Case Study

8.1 Naive vs. Precise Modeling

The "true" effect of the new ad was an 8-second increase in dwell time.

- **Naive Observational**: Effect overestimated (10.68s) because high-dwell-city users chose the new ad.
- **Randomized Study**: Comparing "New" vs "Current" yielded an accurate estimate (8.22s).
- **ANCOVA**: Including confounders (city/age) in a randomized model increases precision (8.04s with a narrower CI).

```
# Observational with Adjustment
lm(y_obs ~ x_self_choice + city + age,
   data = sample_TikTok)

# Randomized Study (ANCOVA)
lm(y_exp ~ x_randomized + city + age,
   data = sample_TikTok)
```

9 Key Takeaways

9.1 Causal Inference

To avoid misleading conclusions, know how data was collected. Randomization allows accurate estimation without knowing every confounder, but including known covariates (ANCOVA) makes estimations more precise.